

Lab 1**Objective:**

The objective of this lab is to simulate the impedance profile of 2 parallel capacitors and estimate the SRF of both.

Method:

ADS was used to simulate the created circuit. An impedance graph was generated with component values of: $C = 10\ \mu\text{F}$ and $1\ \mu\text{F}$, $\text{ESR} = 0.01\ \Omega$, and $\text{ESL} = 3\ \text{nH}$. The SRF is estimated by observing the chart and verified with the f_{SRF} formula: $f_{\text{SRF}} = \frac{1}{2\pi\sqrt{LC}}$.

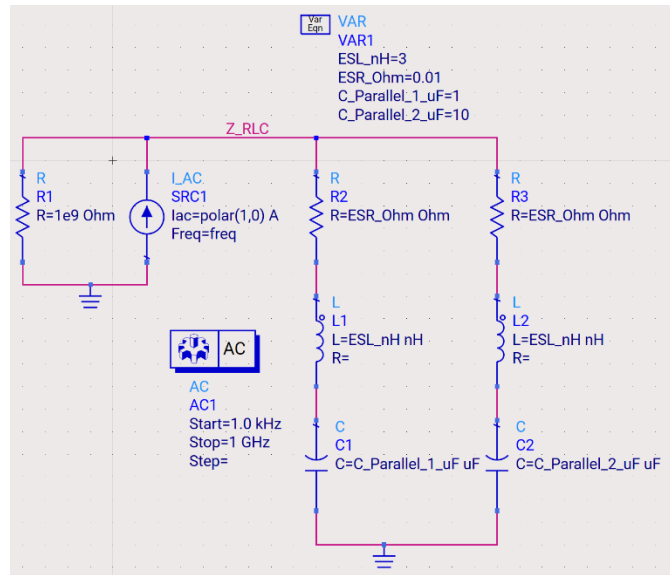


Figure 1: Lab 1 Completed Circuit

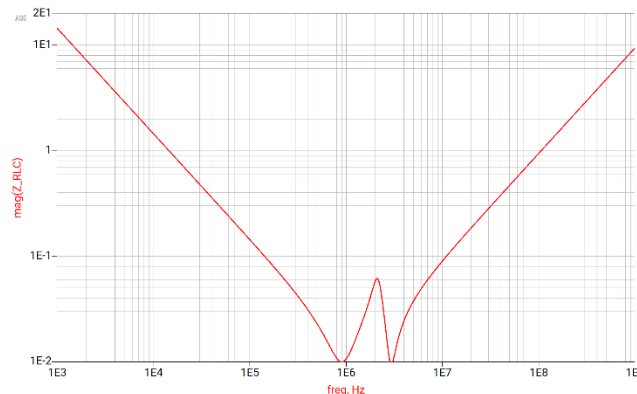
Results and Plots:

Figure 2: Impedance of $1\ \mu\text{F}$ and $10\ \mu\text{F}$ capacitors in parallel

Analysis and Conclusion:

By observing the behavior of the simulated circuit in Figure 2, the estimated SRF value for the $10\ \mu\text{F}$ capacitor (first minimum) is about 1 MHz and the $1\ \mu\text{F}$ capacitor is about 3 MHz. If the component values are entered into the SRF formula the more exact estimates result to $\sim 0.92\ \text{MHz}$ for the $10\ \mu\text{F}$ and $\sim 2.9\ \text{MHz}$ for the $1\ \mu\text{F}$ capacitor.

$$f_{\text{SRF}} = \frac{1}{2\pi\sqrt{(3 \times 10^{-9})(10 \times 10^{-6})}} \approx 0.92\ \text{MHz} \quad f_{\text{SRF}} = \frac{1}{2\pi\sqrt{(3 \times 10^{-9})(10 \times 10^{-6})}} \approx 2.9\ \text{MHz}$$

Based on the results of the simulation it becomes clear that the capacitors behave ideally near lower frequencies and above their SRF, they behave like an inductor. This is evidenced by the decreasing impedance in the beginning and the increasing impedance after the minimums. By using 10 μF and 1 μF it becomes clear that a larger capacitance will cover a lower frequency and a smaller capacitance will cover a high frequency. Adding these in parallel helps cover more frequencies. However, the small peak in the plot between the two SRFs shows that by adding the two capacitors in parallel, they introduce a small spike in impedance that otherwise would not be there with 1 larger capacitor.

This simulation shows that each component value should be chosen carefully as the ESL and ESR value will change where the SRF is and how low it can go. It also shows why some damping can be needed with the unwanted peak in impedance. Without consideration, noise and ringing can be introduced.

Lab 2

Objective:

The objective of this lab is to run multiple simulations of various capacitor circuits and analyze the transient domain and frequency domain behaviors.

Method:

AC and transient simulations were run for 5 different circuits. The first simulates the impedance looking into the capacitor from the die, with an on-die capacitance of 1 nF and an on-die resistance of 1 ohm. The next four are all similar in which a model for a real capacitor is constructed, with the same on-die capacitance and resistance as before, but with two different inductance values of 5 nH and 10 nH, and a different power source. An AC source for the impedance graphs and a switching current of 0.1 A in 1 ns for the transient graphs.

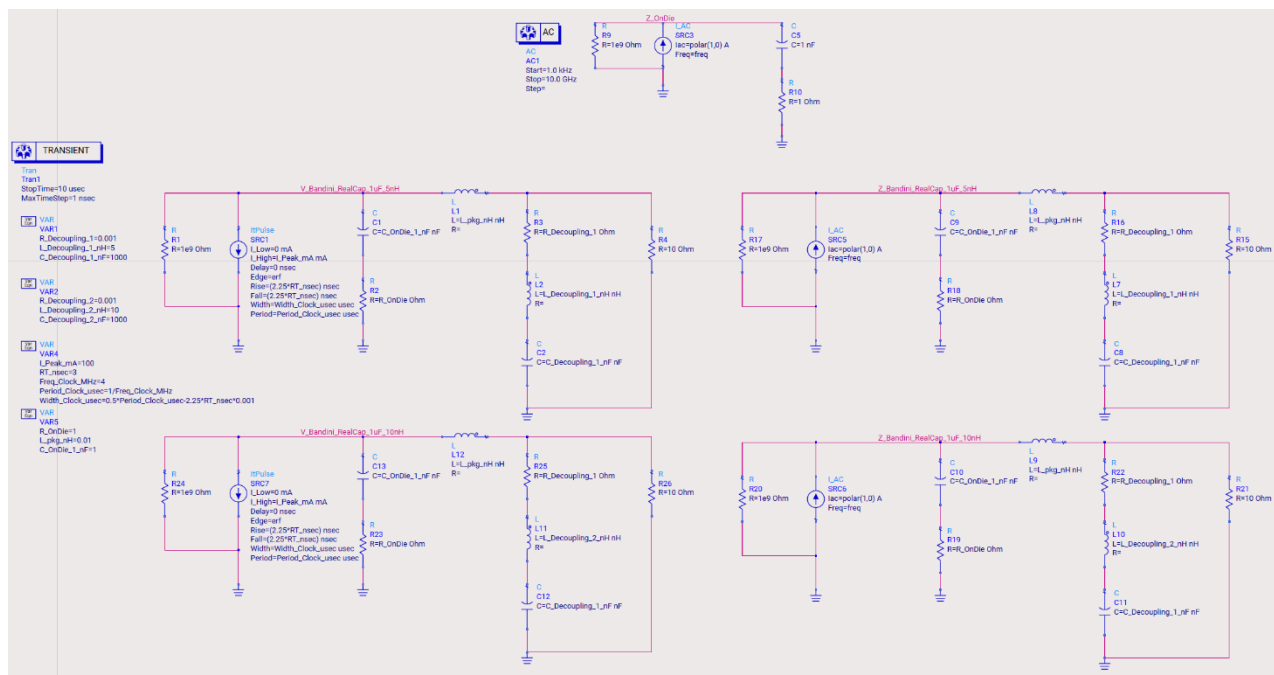


Figure 3: Lab 2 Completed Circuit

Results and Plots:

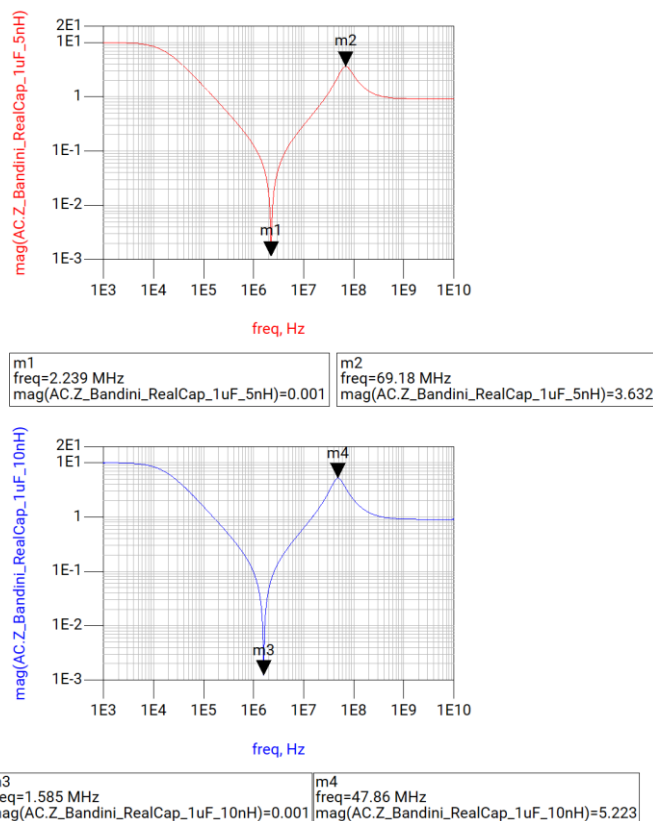


Figure 4: Impedance Magnitude Plots

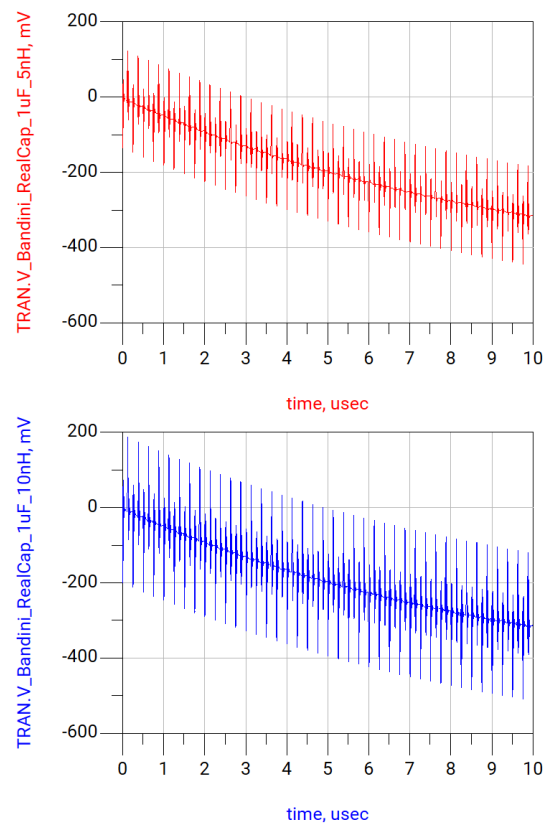


Figure 5: Transient Simulations

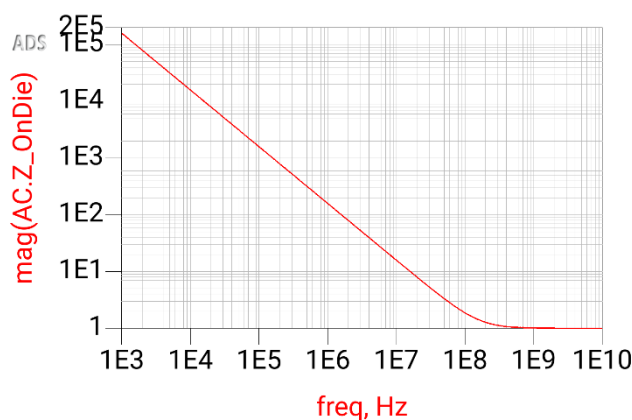


Figure 6: Impedance looking into the capacitor from the die

Analysis and Conclusion:

Figure 6 shows the impedance simulation from looking into the capacitor from the die. It acts as a capacitor at lower frequencies as expected and then flattens out like a resistor at higher frequencies (right where the on-die resistance of $1\ \Omega$ is). With a realistic capacitor model with an off-die decoupling capacitor, there is clear self-resonance effects in the impedance chart. It appears that increasing the ESL value will shift its resonance to a lower frequency. The 10 nH case shows more resonance than the 5 nH case.

The transient simulations show that a higher ESL leads to more ringing and large voltage droop. The 5 nH case shows slightly better performance than the 10 nH circuit. This shows that off-die capacitors, such as this one decoupling capacitor, can be limited by inductance when supplying a transient current.

Combining these results leads to the conclusion that a good decoupling circuit requires minimal inductance with carefully chosen values for each component to ensure minimal parasitic effects.